Harder Induction — Solutions

Warm-up

Question 1a:
We wish to prove that $\sum_{k=1}^{n} \frac{1}{2}k(k+1) = \frac{1}{6}n(n+1)(n+2)$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\text{LHS} = \sum_{k=1}^{1} \frac{1}{2}k(k+1) = \frac{1}{2}(1)(2) = 1$$
$$\text{RHS} = \frac{1}{6}(1)(1+1)(1+2) = \frac{1}{6}(1)(2)(3) = 1$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{k=1}^{k} \frac{1}{2}i(i+1) = \frac{1}{6}k(k+1)(k+2)$$

We must prove it is true for $n = k+1$, i.e., $\sum_{i=1}^{k+1} \frac{1}{2}i(i+1) = \frac{1}{6}(k+1)(k+2)(k+3)$.

$$\begin{aligned}
\sum_{i=1}^{k+1} \frac{1}{2}i(i+1) &= \sum_{i=1}^{k} \frac{1}{2}i(i+1) + \frac{1}{2}(k+1)(k+2) \\
&= \frac{1}{6}k(k+1)(k+2) + \frac{1}{2}(k+1)(k+2) \\
&= (k+1)(k+2)\left[\frac{1}{6}k + \frac{1}{2}\right] \\
&= (k+1)(k+2)\left[\frac{k+3}{6}\right] \\
&= \frac{1}{6}(k+1)(k+2)(k+3)
\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1b:
We wish to prove that $1^2 + 2^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1)$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\text{LHS} = 1^2 = 1$$
$$\text{RHS} = \frac{1}{6}(1)(1+1)(2(1)+1) = \frac{1}{6}(1)(2)(3) = 1$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{i=1}^{k} i^2 = \frac{1}{6}k(k+1)(2k+1)$$

We must prove it is true for $n = k+1$, i.e., $\sum_{i=1}^{k+1} i^2 = \frac{1}{6}(k+1)(k+2)(2(k+1)+1) = \frac{1}{6}(k+1)(k+2)(2k+3)$.

$$\begin{aligned}
\sum_{i=1}^{k+1} i^2 &= \sum_{i=1}^{k} i^2 + (k+1)^2 \\
&= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \\
&= (k+1)\left[\frac{k(2k+1)}{6} + (k+1)\right] \\
&= (k+1)\left[\frac{2k^2+k+6k+6}{6}\right] \\
&= \frac{k+1}{6}(2k^2+7k+6) \\
&= \frac{1}{6}(k+1)(k+2)(2k+3)
\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1c:
Note: This question is a duplicate of 1a. The solution is identical to the solution provided for 1a.

Question 1d:
We wish to prove that $2^n > n$ for $n \in \mathbb{Z}^+$.

Base Case: For $n = 1$,

$$\begin{aligned}
\text{LHS} &= 2^1 = 2 \\
\text{RHS} &= 1
\end{aligned}$$

Since $2 > 1$, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$2^k > k$$

We must prove it is true for $n = k+1$, i.e., $2^{k+1} > k+1$.

$$\begin{aligned}
2^{k+1} &= 2 \cdot 2^k \\
&> 2k \quad (\text{since } 2^k > k) \\
&= k + k
\end{aligned}$$

Since $k \geq 1$, it follows that $k + k \geq k + 1$. Therefore,

$$2^{k+1} > k + 1$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1e:
We wish to prove that $n^2 > 2n + 1$ for $n \geq 3, n \in \mathbb{Z}^+$.

Base Case: For $n = 3$,

$$\begin{aligned}
\text{LHS} &= 3^2 = 9 \\
\text{RHS} &= 2(3) + 1 = 7
\end{aligned}$$

Since $9 > 7$, the statement is true for $n = 3$.
Inductive Step: Assume the statement is true for $n = k$, where $k \geq 3, k \in \mathbb{Z}^+$.

$$k^2 > 2k + 1$$

We must prove it is true for $n = k + 1$, i.e., $(k + 1)^2 > 2(k + 1) + 1 = 2k + 3$.

$$\begin{aligned}(k+1)^2 &= k^2 + 2k + 1 \\ &> (2k + 1) + 2k + 1 \quad (\text{since } k^2 > 2k + 1) \\ &= 4k + 2\end{aligned}$$

Since $k \geq 3$, then $4k + 2 > 2k + 3$ because $2k > 1$, which is true for $k \geq 3$.

$$\therefore (k + 1)^2 > 2k + 3$$

Thus, by the principle of mathematical induction, the statement is true for all $n \geq 3, n \in \mathbb{Z}^+$.

Question 1f:
Given $T_1 = 3$ and $T_n = T_{n-1} + 4$ for $n \geq 2$, we wish to prove $T_n = 4n - 1$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned}\text{LHS} &= T_1 = 3 \\ \text{RHS} &= 4(1) - 1 = 3\end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$T_k = 4k - 1$$

We must prove it is true for $n = k + 1$, i.e., $T_{k+1} = 4(k + 1) - 1 = 4k + 3$.

$$\begin{aligned}T_{k+1} &= T_k + 4 \\ &= (4k - 1) + 4 \\ &= 4k + 3\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1g:
Given $T_1 = 5$ and $T_n = 2T_{n-1} + 1$ for $n \geq 2$, we wish to prove $T_n = 6 \times 2^{n-1} - 1$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned}\text{LHS} &= T_1 = 5 \\ \text{RHS} &= 6 \times 2^{1-1} - 1 = 6(1) - 1 = 5\end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$T_k = 6 \times 2^{k-1} - 1$$

We must prove it is true for $n = k+1$, i.e., $T_{k+1} = 6 \times 2^{(k+1)-1} - 1 = 6 \times 2^k - 1$.

$$\begin{aligned} T_{k+1} &= 2T_k + 1 \\ &= 2(6 \times 2^{k-1} - 1) + 1 \\ &= 6 \times 2^k - 2 + 1 \\ &= 6 \times 2^k - 1 \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1h:
We wish to prove $\frac{d}{dx}x^n = nx^{n-1}$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned} \text{LHS} &= \frac{d}{dx}x^1 = 1 \\ \text{RHS} &= 1x^{1-1} = 1x^0 = 1 \end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\frac{d}{dx}x^k = kx^{k-1}$$

We must prove it is true for $n = k+1$, i.e., $\frac{d}{dx}x^{k+1} = (k+1)x^k$.

$$\begin{aligned} \frac{d}{dx}x^{k+1} &= \frac{d}{dx}(x \cdot x^k) \\ &= x\frac{d}{dx}(x^k) + x^k\frac{d}{dx}(x) \quad \text{(Product Rule)} \\ &= x(kx^{k-1}) + x^k(1) \\ &= kx^k + x^k \\ &= (k+1)x^k \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Skill-building

Question 1a:
We wish to prove that $\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned} \text{LHS} &= 1^3 = 1 \\ \text{RHS} &= \frac{1}{4}(1)^2(1+1)^2 = \frac{1}{4}(1)(4) = 1 \end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$$

We must prove it is true for $n = k+1$, i.e., $\sum_{r=1}^{k+1} r^3 = \frac{1}{4}(k+1)^2(k+2)^2$.

$$\begin{aligned}
\sum_{r=1}^{k+1} r^3 &= \sum_{r=1}^{k} r^3 + (k+1)^3 \\
&= \frac{1}{4}k^2(k+1)^2 + (k+1)^3 \\
&= (k+1)^2\left[\frac{k^2}{4} + (k+1)\right] \\
&= (k+1)^2\left[\frac{k^2+4k+4}{4}\right] \\
&= \frac{1}{4}(k+1)^2(k+2)^2
\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1b:

We wish to prove that $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)} = \frac{n}{2n+1}$ for $n \in \mathbb{Z}^+$.

Base Case: For $n = 1$,

$$\begin{aligned}
\text{LHS} &= \frac{1}{(2(1)-1)(2(1)+1)} = \frac{1}{(1)(3)} = \frac{1}{3} \\
\text{RHS} &= \frac{1}{2(1)+1} = \frac{1}{3}
\end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{k=1}^{k} \frac{1}{(2i-1)(2i+1)} = \frac{k}{2k+1}$$

We must prove it is true for $n = k+1$, i.e., $\sum_{i=1}^{k+1} \frac{1}{(2i-1)(2i+1)} = \frac{k+1}{2(k+1)+1} = \frac{k+1}{2k+3}$.

$$\begin{aligned}
\sum_{i=1}^{k+1} \frac{1}{(2i-1)(2i+1)} &= \sum_{i=1}^{k} \frac{1}{(2i-1)(2i+1)} + \frac{1}{(2(k+1)-1)(2(k+1)+1)} \\
&= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\
&= \frac{k(2k+3)+1}{(2k+1)(2k+3)} \\
&= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\
&= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \\
&= \frac{k+1}{2k+3}
\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1c:
We wish to prove that $n! > n^2$ for $n \geq 4, n \in \mathbb{Z}^+$.
Base Case: For $n = 4$,

$$\text{LHS} = 4! = 24$$
$$\text{RHS} = 4^2 = 16$$

Since $24 > 16$, the statement is true for $n = 4$.
Inductive Step: Assume the statement is true for $n = k$, where $k \geq 4, k \in \mathbb{Z}^+$.

$$k! > k^2$$

We must prove it is true for $n = k + 1$, i.e., $(k+1)! > (k+1)^2$.

$$\begin{aligned}(k+1)! &= (k+1)k! \\ &> (k+1)k^2 \quad (\text{since } k! > k^2) \\ &= k^3 + k^2\end{aligned}$$

We need to show $k^3 + k^2 > (k+1)^2 = k^2 + 2k + 1$. This is equivalent to showing $k^3 > 2k + 1$. Since $k \geq 4$, $k^3 \geq 64$ and $2k + 1 \leq 2(k) + 1$. For $k = 4$, $64 > 9$. Since k^3 grows much faster than $2k + 1$, the inequality holds for all $k \geq 4$.

$$\therefore (k+1)! > (k+1)^2$$

Thus, by the principle of mathematical induction, the statement is true for all $n \geq 4, n \in \mathbb{Z}^+$.

Question 1d:
We wish to prove that $\sum_{i=1}^{n} \frac{1}{i^2} < 2 - \frac{1}{n}$ for $n \geq 2, n \in \mathbb{Z}^+$.
Base Case: For $n = 2$,

$$\text{LHS} = \sum_{i=1}^{2} \frac{1}{i^2} = 1 + \frac{1}{4} = 1.25$$
$$\text{RHS} = 2 - \frac{1}{2} = 1.5$$

Since $1.25 < 1.5$, the statement is true for $n = 2$.
Inductive Step: Assume the statement is true for $n = k$, where $k \geq 2, k \in \mathbb{Z}^+$.

$$\sum_{i=1}^{k} \frac{1}{i^2} < 2 - \frac{1}{k}$$

We must prove it is true for $n = k + 1$, i.e., $\sum_{i=1}^{k+1} \frac{1}{i^2} < 2 - \frac{1}{k+1}$.

$$\begin{aligned}\sum_{i=1}^{k+1} \frac{1}{i^2} &= \sum_{i=1}^{k} \frac{1}{i^2} + \frac{1}{(k+1)^2} \\ &< 2 - \frac{1}{k} + \frac{1}{(k+1)^2}\end{aligned}$$

To complete the proof, we must show that $2-\frac{1}{k}+\frac{1}{(k+1)^2}<2-\frac{1}{k+1}$, which is equivalent to showing:

$$\begin{aligned}\frac{1}{(k+1)^2} &< \frac{1}{k}-\frac{1}{k+1}\\ \frac{1}{(k+1)^2} &< \frac{(k+1)-k}{k(k+1)}\\ \frac{1}{(k+1)^2} &< \frac{1}{k(k+1)}\end{aligned}$$

Since $k(k+1)<(k+1)^2$ for $k\geq 2$, then $\frac{1}{(k+1)^2}<\frac{1}{k(k+1)}$ is true.

$$\therefore \sum_{i=1}^{k+1}\frac{1}{i^2}<2-\frac{1}{k+1}$$

Thus, by the principle of mathematical induction, the statement is true for all $n\geq 2, n\in\mathbb{Z}^+$.

Question 1e:

Given $T_1=a$ and $T_n=T_{n-1}+d$ for $n\geq 2$, we wish to prove $T_n=a+(n-1)d$ for $n\in\mathbb{Z}^+$.

Base Case: For $n=1$,

$$\begin{aligned}\text{LHS} &= T_1=a\\ \text{RHS} &= a+(1-1)d=a\end{aligned}$$

Since LHS = RHS, the statement is true for $n=1$.

Inductive Step: Assume the statement is true for $n=k$, where $k\in\mathbb{Z}^+$.

$$T_k=a+(k-1)d$$

We must prove it is true for $n=k+1$, i.e., $T_{k+1}=a+(k)d$.

$$\begin{aligned}T_{k+1} &= T_k+d\\ &= (a+(k-1)d)+d\\ &= a+kd-d+d\\ &= a+kd\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n\in\mathbb{Z}^+$.

Question 1f:

Given $T_1=a$ and $T_n=rT_{n-1}$ for $n\geq 2$, we wish to prove $T_n=ar^{n-1}$ for $n\in\mathbb{Z}^+$.

Base Case: For $n=1$,

$$\begin{aligned}\text{LHS} &= T_1=a\\ \text{RHS} &= ar^{1-1}=ar^0=a\end{aligned}$$

Since LHS = RHS, the statement is true for $n=1$.

Inductive Step: Assume the statement is true for $n=k$, where $k\in\mathbb{Z}^+$.

$$T_k=ar^{k-1}$$

We must prove it is true for $n = k+1$, i.e., $T_{k+1} = ar^k$.

$$\begin{aligned} T_{k+1} &= rT_k \\ &= r(ar^{k-1}) \\ &= ar^{k-1+1} \\ &= ar^k \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 1g:
Given $\frac{d}{dx}e^x = e^x$, we wish to prove $\frac{d}{dx}e^{nx} = ne^{nx}$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned} \text{LHS} &= \frac{d}{dx}e^{1x} = e^x \\ \text{RHS} &= 1e^{1x} = e^x \end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\frac{d}{dx}e^{kx} = ke^{kx}$$

We must prove it is true for $n = k+1$, i.e., $\frac{d}{dx}e^{(k+1)x} = (k+1)e^{(k+1)x}$.

$$\begin{aligned} \frac{d}{dx}e^{(k+1)x} &= \frac{d}{dx}(e^{kx} \cdot e^x) \\ &= e^{kx}\frac{d}{dx}(e^x) + e^x\frac{d}{dx}(e^{kx}) \quad \text{(Product Rule)} \\ &= e^{kx}(e^x) + e^x(ke^{kx}) \\ &= e^{(k+1)x} + ke^{(k+1)x} \\ &= (k+1)e^{(k+1)x} \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Easier Exam Questions

Question 1:
The provided proof attempts to show that $5^n + 2 \times 11^n$ is a multiple of 3 for all $n \in \mathbb{Z}^+$.
Let's examine the steps: (A) Base case $n = 1$: $5^1 + 2(11)^1 = 5 + 22 = 27$. $27 = 3 \times 9$. Correct. (B) Inductive hypothesis: $5^k + 2 \times 11^k = 3M$. Correct. (C) Inductive step for $n = k+1$:

$$\begin{aligned} 5^{k+1} + 2 \times 11^{k+1} &= 5 \times 5^k + 2 \times 11^k \times 11 \\ &= 5(5^k) + 22(11^k) \end{aligned}$$

In the provided text, step (C) says:

$$= 5(5^k + 2 \times 11^k) + 6 \times 2 \times 11^k$$

Let's check this expansion:

$$\begin{aligned} 5(5^k + 2 \times 11^k) + 6 \times 2 \times 11^k &= 5 \cdot 5^k + 10 \cdot 11^k + 12 \cdot 11^k \\ &= 5 \cdot 5^k + 22 \cdot 11^k \end{aligned}$$

This matches the expression $5^{k+1} + 2 \times 11^{k+1}$. So step (C) is correct.

Now let's look at step (D):

$$\begin{aligned} &= 5(3M) + 3(6 \times 11^k) \\ &= 3(5M + 18 \times 11^k) \end{aligned}$$

Wait, the provided text for (D) says: $= 3(5 + 6 \times 11^k)$. It has omitted the M from the $5(3M)$ term.

$$\text{Provided (D): } 5(3M) + 3(6 \times 11^k) = 3(5 + 6 \times 11^k)$$

This is an algebraic error. The M is missing.

Therefore, the error occurs in section (D).

Question 2:
We wish to prove $u_n = \frac{1}{3}(2^{n+1} + (-1)^n)$ given $u_1 = 1$ and $u_{n+1} = 2u_n + (-1)^{n+1}$.

Base Case: For $n = 1$,

$$\begin{aligned} \text{LHS} &= u_1 = 1 \\ \text{RHS} &= \frac{1}{3}(2^{1+1} + (-1)^1) = \frac{1}{3}(4 - 1) = \frac{3}{3} = 1 \end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$u_k = \frac{1}{3}(2^{k+1} + (-1)^k)$$

We must prove it is true for $n = k + 1$, i.e., $u_{k+1} = \frac{1}{3}(2^{k+2} + (-1)^{k+1})$.

$$\begin{aligned} u_{k+1} &= 2u_k + (-1)^{k+1} \\ &= 2\left[\frac{1}{3}(2^{k+1} + (-1)^k)\right] + (-1)^{k+1} \\ &= \frac{2 \cdot 2^{k+1} + 2(-1)^k}{3} + (-1)^{k+1} \\ &= \frac{2^{k+2} + 2(-1)^k + 3(-1)^{k+1}}{3} \\ &= \frac{2^{k+2} - 2(-1)^{k+1} + 3(-1)^{k+1}}{3} \quad (\text{since } (-1)^k = -(-1)^{k+1}) \\ &= \frac{2^{k+2} + (-1)^{k+1}}{3} \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 3:
We wish to prove $\sum_{i=1}^{n} \sqrt{i} > \frac{2n\sqrt{n}}{3}$ for $n \in \mathbb{Z}^+$.

Base Case: For $n = 1$,

$$\text{LHS} = \sqrt{1} = 1$$
$$\text{RHS} = \frac{2(1)\sqrt{1}}{3} = \frac{2}{3}$$

Since $1 > \frac{2}{3}$, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{i=1}^{k} \sqrt{i} > \frac{2k\sqrt{k}}{3}$$

We must prove it is true for $n = k+1$, i.e., $\sum_{i=1}^{k+1} \sqrt{i} > \frac{2(k+1)\sqrt{k+1}}{3}$.

$$\begin{aligned}\sum_{i=1}^{k+1} \sqrt{i} &= \sum_{i=1}^{k} \sqrt{i} + \sqrt{k+1} \\ &> \frac{2k\sqrt{k}}{3} + \sqrt{k+1}\end{aligned}$$

We need to show that $\frac{2k\sqrt{k}}{3} + \sqrt{k+1} > \frac{2(k+1)\sqrt{k+1}}{3}$. This is equivalent to showing $\sqrt{k+1} - \frac{2(k+1)\sqrt{k+1}}{3} > -\frac{2k\sqrt{k}}{3}$, or

$$\begin{aligned}\frac{3\sqrt{k+1} - 2(k+1)\sqrt{k+1}}{3} &> -\frac{2k\sqrt{k}}{3} \\ \frac{(3-2k-2)\sqrt{k+1}}{3} &> -\frac{2k\sqrt{k}}{3} \\ \frac{(1-2k)\sqrt{k+1}}{3} &> -\frac{2k\sqrt{k}}{3}\end{aligned}$$

Multiplying by 3 and rearranging:

$$2k\sqrt{k} > (2k-1)\sqrt{k+1}$$

Squaring both sides (since both are positive for $k \geq 1$):

$$\begin{aligned}4k^3 &> (2k-1)^2(k+1) \\ 4k^3 &> (4k^2-4k+1)(k+1) \\ 4k^3 &> 4k^3+4k^2-4k^2-4k+k+1 \\ 4k^3 &> 4k^3-3k+1\end{aligned}$$

Since $3k > 1$ for $k \geq 1$, the inequality $4k^3 > 4k^3 - 3k + 1$ is true.

$$\therefore \sum_{i=1}^{k+1} \sqrt{i} > \frac{2(k+1)\sqrt{k+1}}{3}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 4:
We wish to prove $n! < (\frac{n}{2})^n$ for $n > 5, n \in \mathbb{N}$.
Base Case: For $n = 6$,

$$\text{LHS} = 6! = 720$$
$$\text{RHS} = (\frac{6}{2})^6 = 3^6 = 729$$

Since $720 < 729$, the statement is true for $n = 6$.
Inductive Step: Assume the statement is true for $n = k$, where $k > 5, k \in \mathbb{N}$.

$$k! < (\frac{k}{2})^k$$

We must prove it is true for $n = k + 1$, i.e., $(k + 1)! < (\frac{k+1}{2})^{k+1}$.

$$\begin{aligned} (k+1)! &= (k+1)k! \\ &< (k+1)\left(\frac{k}{2}\right)^k \quad (\text{since } k! < (\frac{k}{2})^k) \\ &= \frac{(k+1)k^k}{2^k} \end{aligned}$$

We need to show that $\frac{(k+1)k^k}{2^k} < \frac{(k+1)^{k+1}}{2^{k+1}}$. This is equivalent to showing:

$$\begin{aligned} \frac{(k+1)k^k}{2^k} &< \frac{(k+1)(k+1)^k}{2 \cdot 2^k} \\ \frac{k^k}{1} &< \frac{(k+1)^k}{2} \\ 2k^k &< (k+1)^k \\ 2 < \left(\frac{k+1}{k}\right)^k &= (1 + \frac{1}{k})^k \end{aligned}$$

We know that the sequence $(1 + \frac{1}{k})^k$ is increasing and approaches $e \approx 2.718$ as $k \to \infty$. For $k = 6$, $(1 + \frac{1}{6})^6 = (\frac{7}{6})^6 \approx 2.52$. Since $2.52 > 2$, the inequality holds for all $k \geq 6$.

$$\therefore (k+1)! < (\frac{k+1}{2})^{k+1}$$

Thus, by the principle of mathematical induction, the statement is true for all $n > 5, n \in \mathbb{N}$.

Question 5:
We wish to prove $\sum_{r=1}^{n} \frac{r}{(r+1)!} = 1 - \frac{1}{(n+1)!}$ for $n \geq 1, n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\text{LHS} = \frac{1}{(1+1)!} = \frac{1}{2}$$
$$\text{RHS} = 1 - \frac{1}{(1+1)!} = 1 - \frac{1}{2} = \frac{1}{2}$$

Since LHS = RHS, the statement is true for $n = 1$.

Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$\sum_{r=1}^{k} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+1)!}$$

We must prove it is true for $n = k+1$, i.e., $\sum_{r=1}^{k+1} \frac{r}{(r+1)!} = 1 - \frac{1}{(k+2)!}$.

$$\begin{aligned}
\sum_{r=1}^{k+1} \frac{r}{(r+1)!} &= \sum_{r=1}^{k} \frac{r}{(r+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \\
&= 1 - \left[\frac{1}{(k+1)!} - \frac{k+1}{(k+2)!}\right] \\
&= 1 - \left[\frac{k+2}{(k+2)!} - \frac{k+1}{(k+2)!}\right] \\
&= 1 - \left[\frac{(k+2)-(k+1)}{(k+2)!}\right] \\
&= 1 - \frac{1}{(k+2)!}
\end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \geq 1, n \in \mathbb{Z}^+$.

Harder Exam Questions

Question 1:

We are given that the product of n consecutive numbers is divisible by $n!$. We wish to prove that $n^5 - n$ is divisible by 2, 3, and 5 for $n \geq 2, n \in \mathbb{Z}^+$.

First, let's factor $n^5 - n$:

$$\begin{aligned}
n^5 - n &= n(n^4 - 1) \\
&= n(n^2 - 1)(n^2 + 1) \\
&= n(n-1)(n+1)(n^2 + 1)
\end{aligned}$$

Divisibility by 2: $n(n-1)$ is the product of two consecutive integers, so it is always divisible by $2! = 2$. Thus $n^5 - n$ is divisible by 2.

Divisibility by 3: $(n-1)n(n+1)$ is the product of three consecutive integers, so it is always divisible by $3! = 6$. Since it is divisible by 6, it is also divisible by 3. Thus $n^5 - n$ is divisible by 3.

Divisibility by 5: We use induction to prove $n^5 - n$ is divisible by 5 for $n \geq 2$.

Base Case: For $n = 2$,

$$2^5 - 2 = 32 - 2 = 30$$

Since $30 = 5 \times 6$, it is divisible by 5. True for $n = 2$.

Inductive Step: Assume $k^5 - k$ is divisible by 5, i.e., $k^5 - k = 5M$ for $M \in \mathbb{Z}$.

We must prove $(k+1)^5 - (k+1)$ is divisible by 5.

$$\begin{aligned}(k+1)^5-(k+1) &= (k^5+5k^4+10k^3+10k^2+5k+1)-(k+1)\\ &= (k^5-k)+5k^4+10k^3+10k^2+5k\\ &= 5M+5(k^4+2k^3+2k^2+k)\\ &= 5(M+k^4+2k^3+2k^2+k)\end{aligned}$$

Since the expression is a multiple of 5, it is divisible by 5. Thus, n^5-n is divisible by 2, 3, and 5 for all $n \geq 2$.

Question 2:
We wish to prove that $7^n+13^n+19^n$ is divisible by 13 for all positive odd integers n.
Let $n=2k-1$ for $k \in \mathbb{Z}^+$.
Base Case: For $k=1$ (so $n=1$),

$$\text{LHS} = 7^1+13^1+19^1 = 7+13+19 = 39$$

Since $39 = 13 \times 3$, it is divisible by 13. True for $n=1$.
Inductive Step: Assume the statement is true for $n=m$, where m is a positive odd integer.

$$7^m+13^m+19^m = 13M \quad (M \in \mathbb{Z})$$

We must prove it is true for the next odd integer $n=m+2$.

$$\begin{aligned}7^{m+2}+13^{m+2}+19^{m+2} &= 7^2\cdot 7^m+13^2\cdot 13^m+19^2\cdot 19^m\\ &= 49\cdot 7^m+169\cdot 13^m+361\cdot 19^m\\ &= 49(7^m)+49(13^m)+120(13^m)+49(19^m)+312(19^m)\\ &= 49(7^m+13^m+19^m)+13\cdot 120(13^{m-1})+13\cdot 24(19^m)\\ &= 49(13M)+13\cdot 120(13^{m-1})+13\cdot 24(19^m)\\ &= 13(49M+120(13^{m-1})+24(19^m))\end{aligned}$$

Which is divisible by 13. Thus $7^{m+2}+13^{m+2}+19^{m+2}$ is divisible by 13. Therefore, by mathematical induction, $7^n+13^n+19^n$ is divisible by 13 for all positive odd integers n.

Question 3:
We wish to prove $T_n = 2\cos\left(\frac{\pi}{2^{n+1}}\right)$ for $n \geq 0, n \in \mathbb{Z}$.
Base Case: For $n=0$,

$$\begin{aligned}\text{LHS} &= T_0 = 0\\ \text{RHS} &= 2\cos\left(\frac{\pi}{2^{0+1}}\right) = 2\cos\left(\frac{\pi}{2}\right) = 2(0) = 0\end{aligned}$$

Since LHS = RHS, the statement is true for $n=0$.
Inductive Step: Assume the statement is true for $n=k$, where $k \geq 0, k \in \mathbb{Z}$.

$$T_k = 2\cos\left(\frac{\pi}{2^{k+1}}\right)$$

We must prove it is true for $n = k + 1$, i.e., $T_{k+1} = 2\cos\left(\frac{\pi}{2^{k+2}}\right)$.

$$\begin{aligned} T_{k+1} &= \sqrt{T_k + 2} \\ &= \sqrt{2\cos\left(\frac{\pi}{2^{k+1}}\right) + 2} \\ &= \sqrt{2\left(1 + \cos\left(\frac{\pi}{2^{k+1}}\right)\right)} \end{aligned}$$

Using the half-angle identity $\cos(2\theta) = 2\cos^2(\theta) - 1$, we have $1 + \cos(2\theta) = 2\cos^2(\theta)$. Let $2\theta = \frac{\pi}{2^{k+1}}$, so $\theta = \frac{\pi}{2^{k+2}}$.

$$\begin{aligned} T_{k+1} &= \sqrt{2\left(2\cos^2\left(\frac{\pi}{2^{k+2}}\right)\right)} \\ &= \sqrt{4\cos^2\left(\frac{\pi}{2^{k+2}}\right)} \\ &= 2\cos\left(\frac{\pi}{2^{k+2}}\right) \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \geq 0, n \in \mathbb{Z}$.

Question 4:
Given $u_1 = 7$ and $u_{n+1} = 2u_n + 3$, we wish to prove $u_n = 5(2^n) - 3$ for $n \in \mathbb{Z}^+$.
Base Case: For $n = 1$,

$$\begin{aligned} \text{LHS} &= u_1 = 7 \\ \text{RHS} &= 5(2^1) - 3 = 10 - 3 = 7 \end{aligned}$$

Since LHS = RHS, the statement is true for $n = 1$.
Inductive Step: Assume the statement is true for $n = k$, where $k \in \mathbb{Z}^+$.

$$u_k = 5(2^k) - 3$$

We must prove it is true for $n = k + 1$, i.e., $u_{k+1} = 5(2^{k+1}) - 3$.

$$\begin{aligned} u_{k+1} &= 2u_k + 3 \\ &= 2(5(2^k) - 3) + 3 \\ &= 10(2^k) - 6 + 3 \\ &= 5(2 \cdot 2^k) - 3 \\ &= 5(2^{k+1}) - 3 \end{aligned}$$

Thus, by the principle of mathematical induction, the statement is true for all $n \in \mathbb{Z}^+$.

Question 5:
We wish to prove that $4^{n+1} + 6^n$ is divisible by 10 when n is even. Let $n = 2k$ for $k \in \mathbb{Z}^+$.
Base Case: For $k = 1$ (so $n = 2$),

$$\text{LHS} = 4^{2+1} + 6^2 = 4^3 + 36 = 64 + 36 = 100$$

Since $100 = 10 \times 10$, it is divisible by 10. True for $n = 2$.

Inductive Step: Assume the statement is true for $n = 2k$, where $k \in \mathbb{Z}^+$.

$$4^{2k+1} + 6^{2k} = 10M \quad (M \in \mathbb{Z})$$

We must prove it is true for the next even integer $n = 2k + 2$.

$$\begin{aligned}
4^{(2k+2)+1} + 6^{2k+2} &= 4^{2k+3} + 6^{2k+2} \\
&= 4^2 \cdot 4^{2k+1} + 6^2 \cdot 6^{2k} \\
&= 16 \cdot 4^{2k+1} + 36 \cdot 6^{2k} \\
&= 16(4^{2k+1} + 6^{2k}) - 16(6^{2k}) + 36(6^{2k}) \\
&= 16(10M) + 20(6^{2k}) \\
&= 10(16M + 2 \cdot 6^{2k})
\end{aligned}$$

Since the expression is a multiple of 10, it is divisible by 10. Thus, $4^{n+1} + 6^n$ is divisible by 10 when n is even.